
MODEL SELECTION AND VALIDATION REPORT (general_kde_fric model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (general_kde_fric model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: student(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	-0.15	0.10
sbt_mean	0.15	0.04
sbt_sd	-0.01	0.03
sst_sd	0.05	0.02
sic_mean	-0.03	0.04
log_chla_mean	0.00	0.02
depth	-0.12	0.02

Dispersion parameter: 0.57

Spatiotemporal AR1 correlation (rho): 0.92

Matérn range: 234.45

Spatiotemporal marginal AR1 SD: 0.50

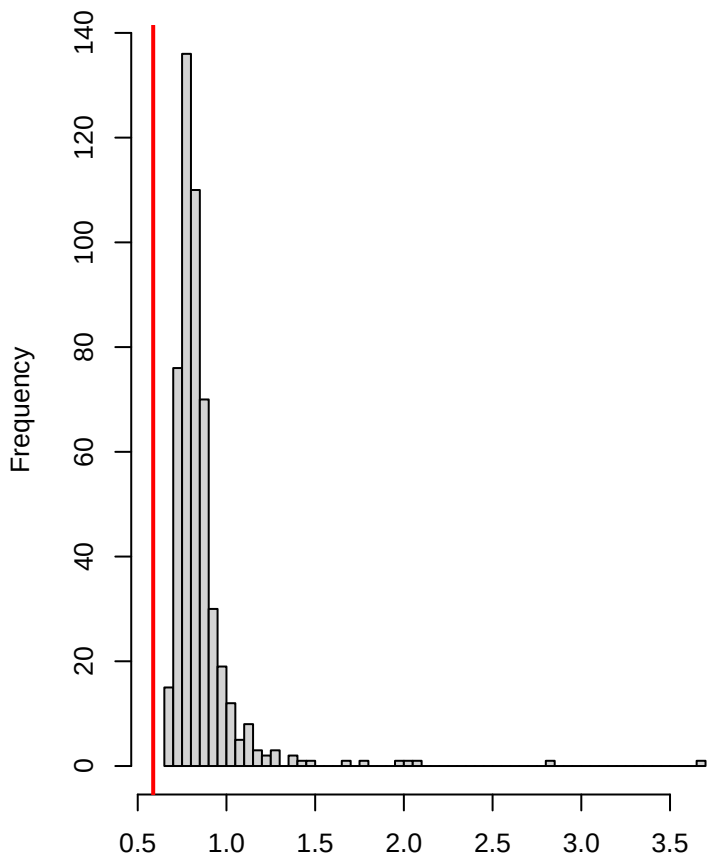
REML criterion at convergence: 4255.155

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

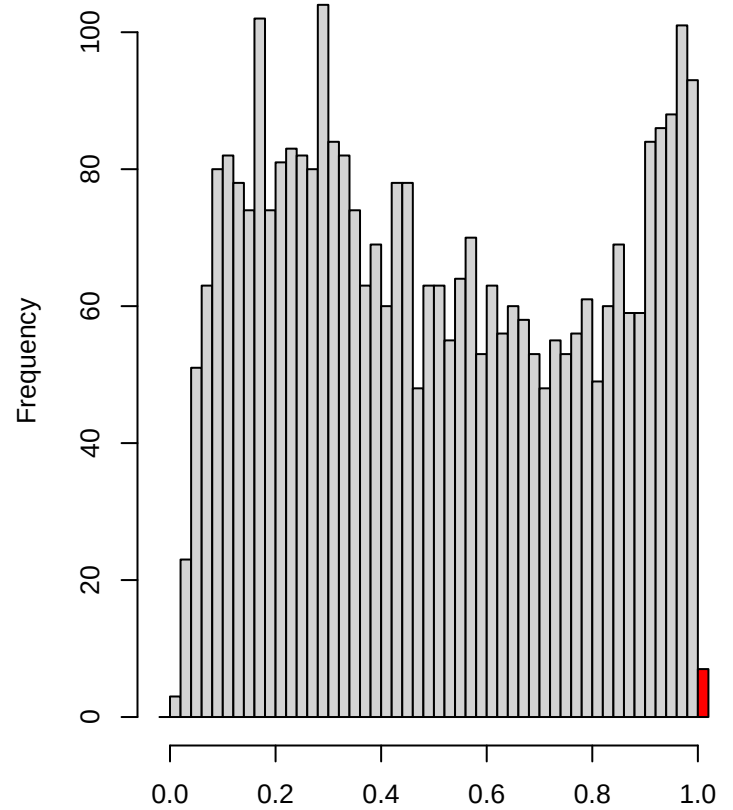
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	11	8532.310	3.298312
spatiotemporal_random_fields_ar1_with_anisotropy	13	8535.609	77.665653
spatial_random_fields	10	8613.274	3.816176
spatial_random_fields_with_anisotropy	12	8617.090	114.522132
spatiotemporal_random_fields_iid	10	8731.613	3.555108
spatiotemporal_random_fields_iid_with_anisotropy	12	8735.168	192.226237
error_only	8	8927.394	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



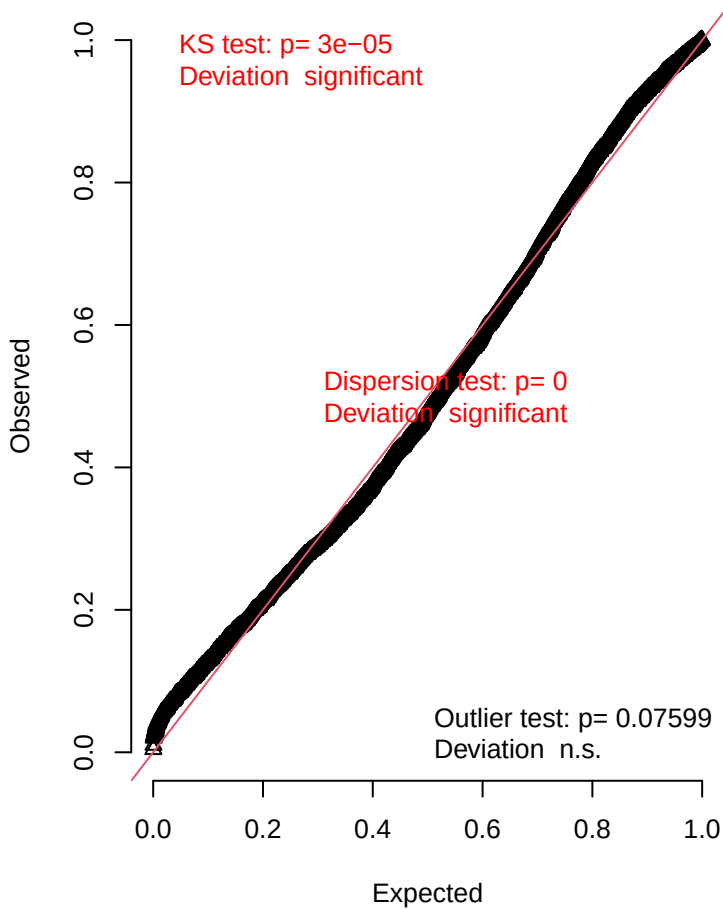
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test n.s.

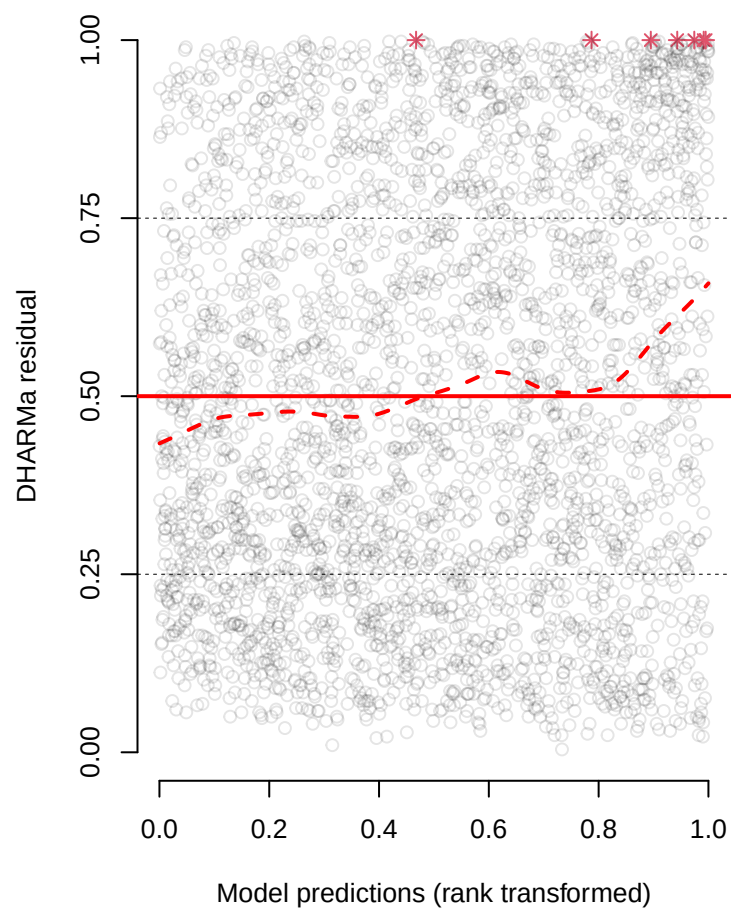


Residuals (outliers are marked red)

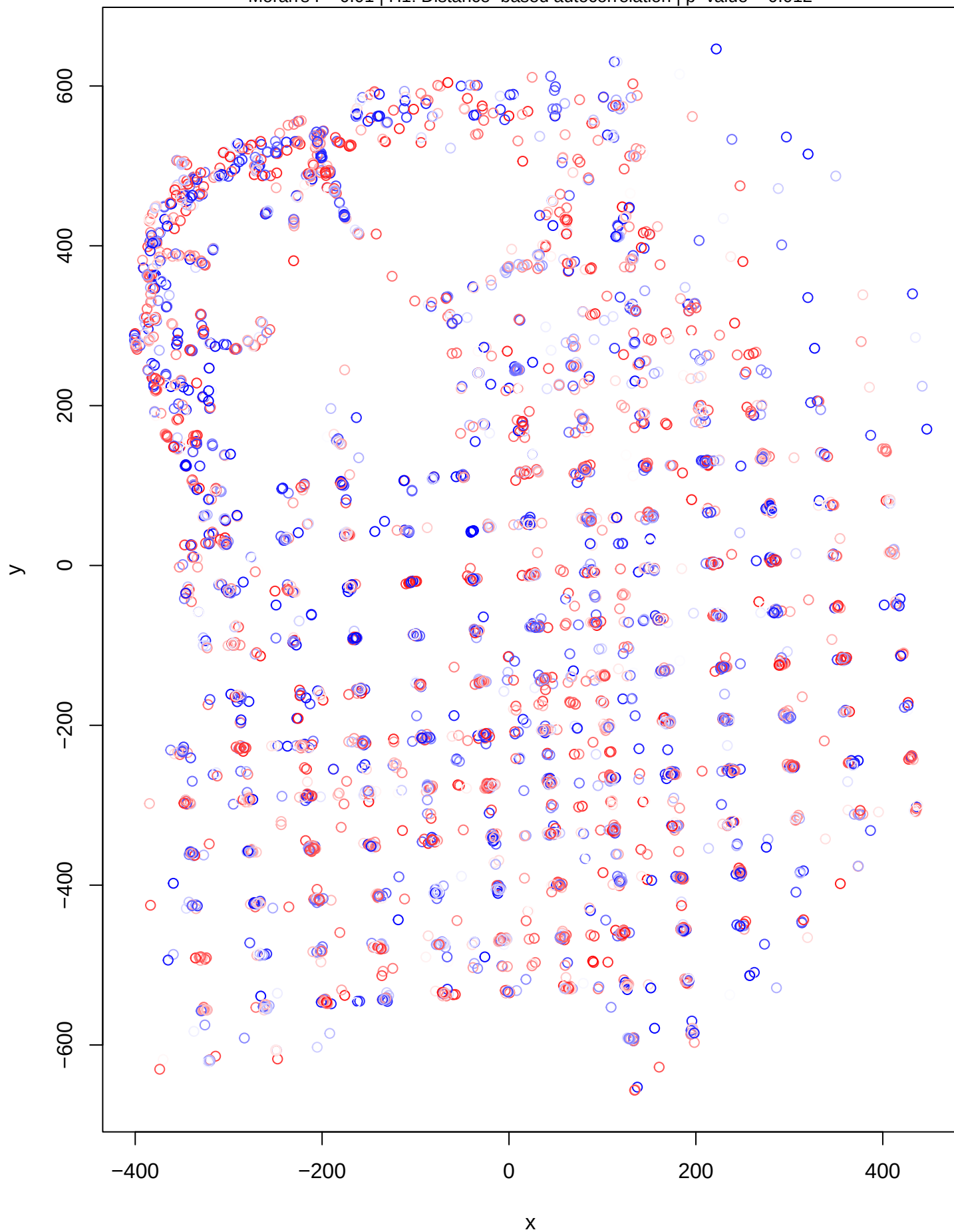
QQ plot residuals



DHARMa residual vs. predicted

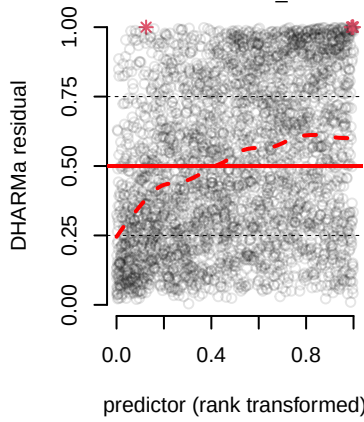


DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.01 | H1: Distance-based autocorrelation | p-value = 0.012



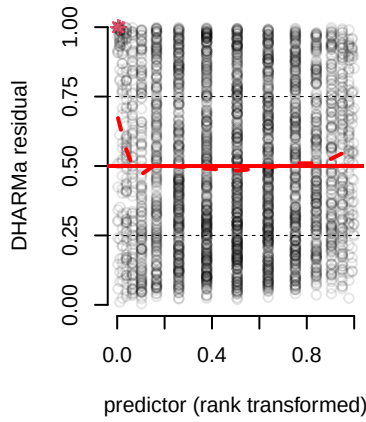
DHARMa residual Residual vs. pred

Predictor: kde_feve



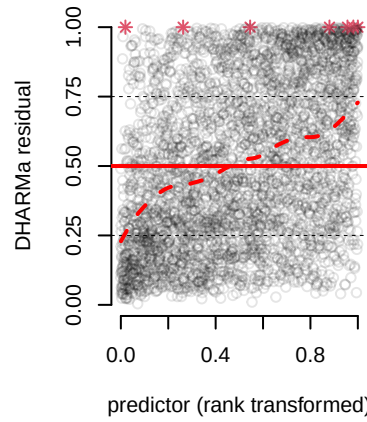
DHARMa residual Residual vs. pred

Predictor: tric



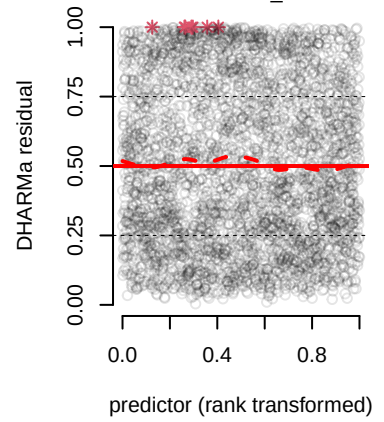
DHARMa residual Residual vs. pred

Predictor: teve



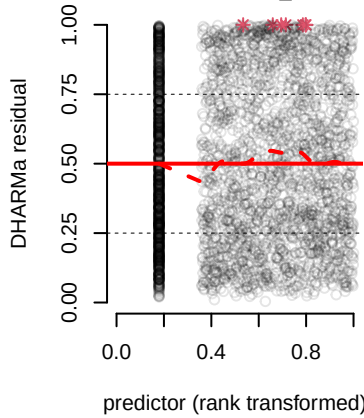
DHARMa residual Residual vs. pred

Predictor: sst_mean



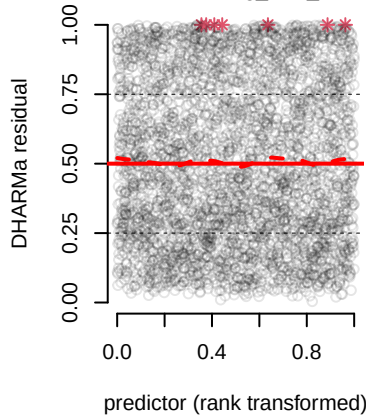
DHARMa residual Residual vs. pred

Predictor: sic_sd



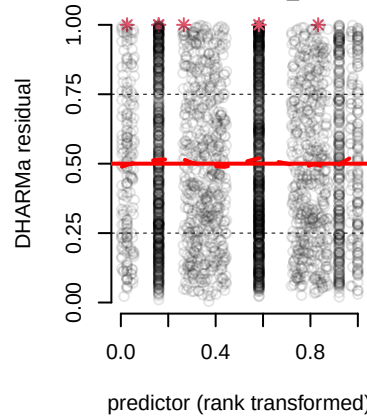
DHARMa residual Residual vs. pred

Predictor: log_chla_sd



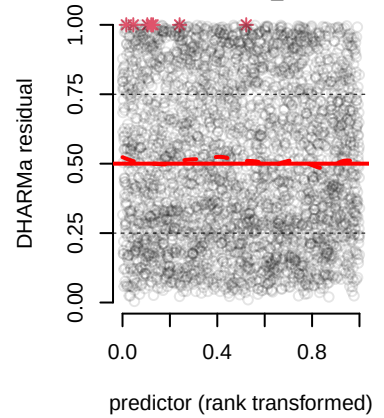
DHARMa residual Residual vs. pred

Predictor: haul_dur



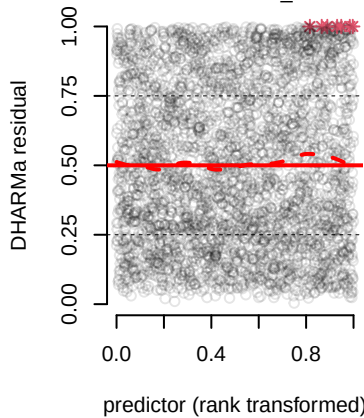
DHARMa residual Residual vs. pred

Predictor: sbt_mean



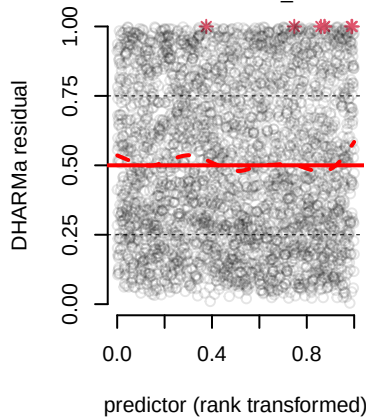
DHARMa residual Residual vs. pred

Predictor: sbt_sd



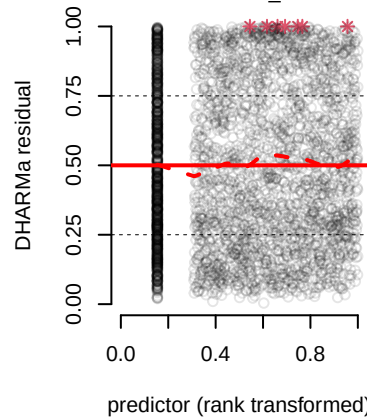
DHARMa residual Residual vs. pred

Predictor: sst_sd



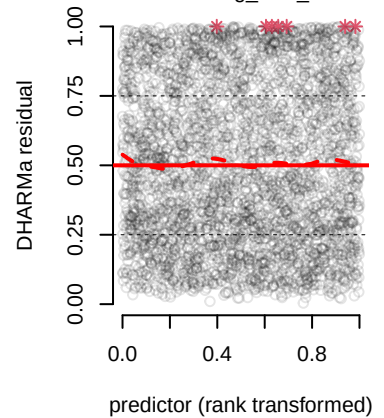
DHARMa residual Residual vs. pred

Predictor: sic_mean



DHARMa residual Residual vs. pred

Predictor: log_chla_mean



DHARMa residual Residual vs. pred

Predictor: depth

